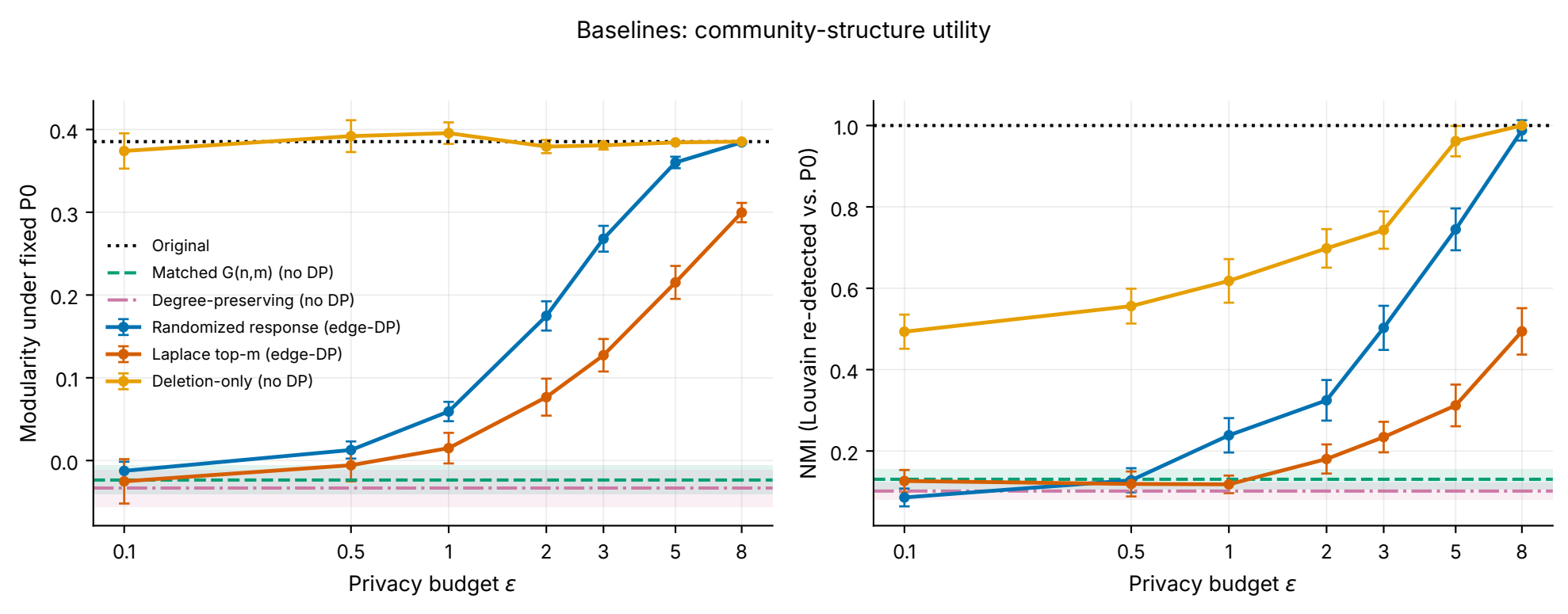
Baselines: community-structure utility
Modularity under fixed P0
0.4
0.3
0.2
0.1
0.0
Original
Matched G(n,m) (no DP)
Degree-preserving (no DP)
Randomized response (edge-DP)
Laplace top-m (edge-DP)
Deletion-only (no DP)
0.1
0.5
1
2
3
5
8
Privacy budget ε
NMI (Louvain re-detected vs. P0)
1.0
0.8
0.6
0.4
0.2
0.1
0.5
1
2
3
5
8
Privacy budget ε